

Image courtesy: www.hackaday.com

uECG

This device helps track the health and heart data of a person. Unlike other ECG devices, this open source device is very

small, weighs just nine grams, and is quite affordable. It gives a backup of up to nine hours and you can obtain the recorded data on both phone and PC.

For details: <https://hackaday.io/project/164486-uecg-a-very-small-wearable-ecg>
<https://github.com/ultimaterobotics/uECG>

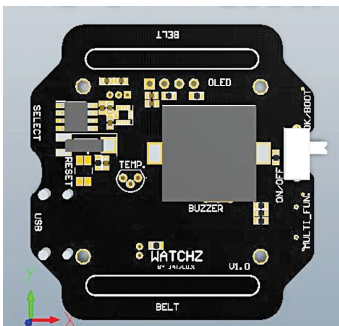


Thermal screening camera with recognition

While other contactless temperature sensors and thermometers need a person to check the results one by one, which can be tiring, this device combines facial recognition

features and an automatic contactless thermal screening system. It can be deployed at entrance to places like offices, malls, or airports. It includes a robust database that stores images of regular visitors' faces along with their names and temperature at the time of entrance for future tracking.

For details: <https://www.electronicshobby.com/electronic-projects/smart-door-camera-with-facial-recognition-feature-for-thermal-screening>



Smart Watchz

This Smart Watchz uses a sensor for capturing a person's temperature, hand movements for sneeze and cough detection, and pulse and SpO2 meters for health data. Using the collected data, the system tries to detect the Covid-19

symptoms. It also supports data logging.

For details: <https://www.instructables.com/Smart-Watchz-With-Corona-Symptoms-Detection-and-Data-Logging/>



Ninja Lamp PCR

Besides being cost-effective and affordable, the Loop-Mediated-Isothermal-Amplification

(Lamp) PCR is easy to make. You just need a few resistors, MOSFETs, heater for tube holder, and connect them to an Arduino. The device gets ready in no time at a cost of about 300 dollars.

For Arduino code and details: <https://hackaday.io/project/174501-covid-19-detectors-300-real-time-pcr-50-lamp>
<https://github.com/hisashin/NinjaLAMP>



Skin temperature scanner

While thermal screening cameras are quite costly, this open source temperature scanning device enables you to make a thermal camera at an exceptionally low cost. The whole device is also small

and portable; it can be carried and mounted anywhere.

For details: <https://hackaday.io/project/170595-skin-temperature-scanner>



Pulmonary ventilator

This ventilator helps a person facing difficulty in breathing. It has a face mask that pushes the pressurised

air through a patient's nose or mouth for breathing. It is an open source non-invasive ventilator that can be tried in an emergency, as long as the patient is not sedated or intubated.

For details: <https://create.arduino.cc/projecthub/david-pascoal/open-source-covid-19-pulmonary-ventilator-4f4586>

By: Ashwini Kumar Sinha

The author is a technical journalist and DIY enthusiast at EFY.

The article was originally published in the June 2021 issue of Electronics For You.